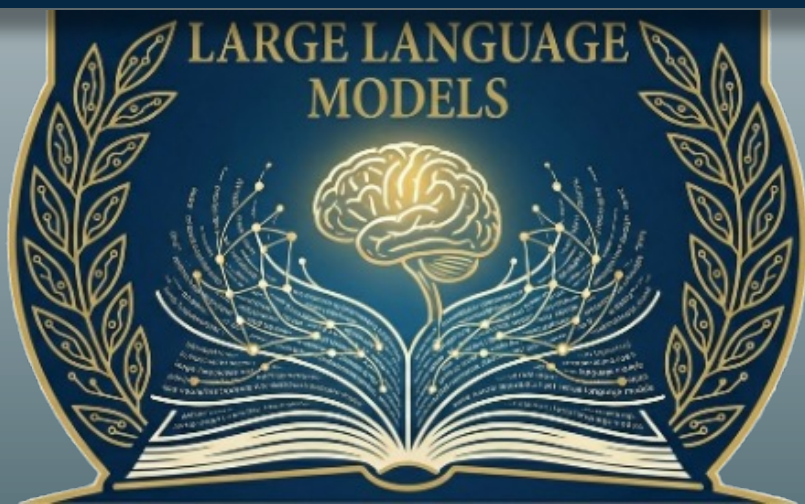


Understanding LLMs

Exploring External and Internal Perspectives



Mohammad Javad Dousti

Source:

* Dr. Yaghoubzadeh's slides

Understanding LLMs

- **Why** do LLMs generate **what** they generate and **how** do they do that!

Two views

- ❑ External mind-like
- ❑ Internal mechanisms

Gemini

2.0 Flash ▼

✦ Try Gemini Advanced



یک بیت زیبا مناسب هوای بهاری اردیبهشت میخوام



باغ پر از شکوفه و بستان پر از گل است
وین بلبل از ترانه، دایم به شور و غلغل است

معنی: باغ پر از شکوفه‌های زیبا و بستان پر از گل‌های رنگارنگ شده است و این بلبل عاشق نیز با آوازهای مست خود، شور و هیجانی دائمی به پا کرده است.



میخواهم سر کلاس درس بگم مدلهای
زبانی چقدر باهوشن. مدلایی مثل
خودت. خوشحالی؟

خیلی! البته اگر منظورت اینه که به دانشجوها
نشون بدی که چقدر می‌تونن خوب زبان رو درک
و تولید کنن، واقعاً موضوع جذابییه. می‌تونن مثلاً
بگی:

«این مدل‌ها میلیاردها کلمه خوندن، مثل یه
دانشجوی شب امتحانی که هیچ‌وقت خسته
نمی‌شه، و حالا می‌تونن براتون مقاله بنویسن،
شعر بگن، برنامه‌نویسی کنن و حتی درباره
احساسات حرف بزنن!»

LLMs as “mind”

- ❑ **Anthropomorphism:** We naturally attribute human-like qualities (knowledge, intent, understanding) to LLMs.

- Driven by their performance on diverse, complex tasks.



- ❑ Cognitive Science: **Theory of Mind**

- Humans are wired to infer mental states in others (and even objects!).
- It is called a "theory" because we cannot directly observe someone else's mind.
- Even LLMs can show some levels of ToM.

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میخوام توضیح بدی که چطور به همچین جوابی رسیدی؟

عالیه، بذار قدم به قدم توضیح بدم که چطور به این پاسخ رسیدم:

۱. تحلیل نیت تو

تو گفتی:

"میخوام سر کلاس درس بگم مدل های زبانی چقدر باهوشن. مدلاپی مثل خودت. خوشحالی؟"

از این جمله متوجه شدم:

- می‌خوای در کلاس یه بحث انگیزشی یا جالب درباره زبان مدل‌ها (مثل من!) راه بندازی.
- حالت غیررسمی و خودمونی صحبت کردی، پس احتمالاً دوست داری این بحث رو صمیمی و قابل فهم ارائه بدی، نه خشک و آکادمیک.
- پرسیدی "خوشحالی؟"، که یه جور شوخی یا همدلی بود، پس خواستم منم در ادامه با همون لحن گرم و شوخ طبع جواب بدم، ولی هم‌زمان آموزنده.

۲. هدف پاسخ

می‌خواستم هم:

- حس کنجکاوی و هیجان دانشجوها رو برانگیزم (چون این چیزیه که تو احتمالاً سر کلاس می‌خوای)،
- هم چندتا ویژگی واقعی مدل‌های زبانی رو توی یه فرم قابل لمس نشون بدم.

برای همین گفتم:

«این مدل‌ها میلیاردها کلمه خوندن، مثل یه دانشجوی شب امتحانی که هیچ‌وقت خسته نمی‌شه...»

→ چون این تشبیه هم طنز داره، هم به خوبی نشون می‌ده که مدل‌ها چقدر زیاد «مطالعه» کردن.

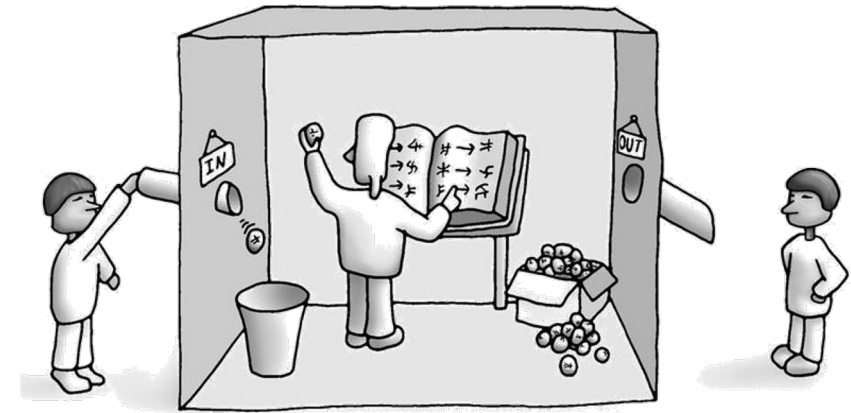
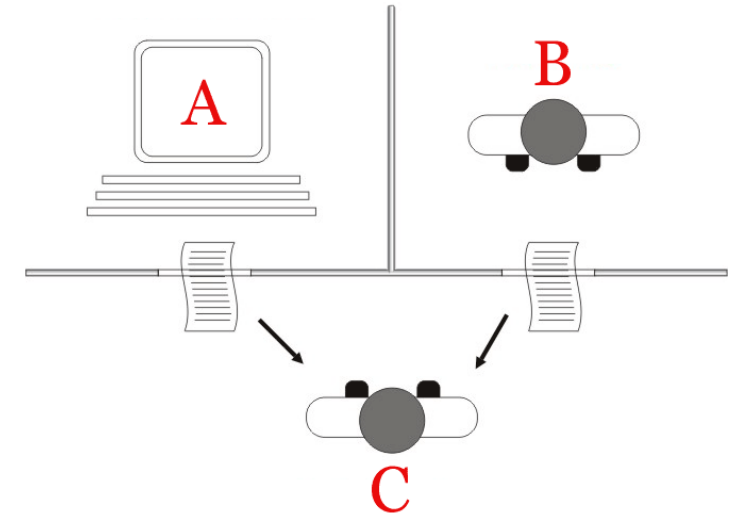
→ بعد گفتم چه کارایی می‌تونن بکنن: «مقاله بنویسن، شعر بگن، برنامه‌نویسی کنن، درباره احساسات حرف بزنن»، تا نشان بدم که «باهوش بودن» یعنی چی در این زمینه.

۳. تنظیم لحن پاسخ

- چون تو با لحنی دوستانه و غیررسمی حرف زدی، منم جواب رو با همون لحن نوشتم.
- خواستم حس خوبی از تعامل با مدل داشته باشی و این حس رو بتونی به دانشجوها هم منتقل کنی.

The Illusion of Understanding?

- ❑ “Turing (1950) proposed what is now known as the **Turing Test**: if a computer could pass for human in on-line chat, it should be counted as **intelligent**.”
- ❑ **Chinese Room Argument**: Manipulating symbols vs. understanding **meaning**.
 - “In 1980, U.C. Berkeley philosopher John Searle introduced a short and widely-discussed argument intended to show conclusively that it is impossible for digital computers to understand language or think, now or in the future”



John Searle's Chinese room argument

- ❑ Imagine a room with a person inside who **does not know a single word of Chinese**.
- ❑ **The Input:** People outside the room slide pieces of paper with Chinese characters written on them through a slot.
- ❑ **The Rulebook:** Inside the room, the person has a massive English rulebook (an algorithm) that tells them exactly how to respond: *"If you see shape X, write down shape Y."*
- ❑ **The Output:** Following these instructions, the person produces perfect Chinese responses and slides them back out.
- ❑ To the people outside, it appears the person inside speaks fluent Chinese. But in reality, the person is just manipulating symbols without understanding their meaning.

Techniques for understanding LLMs as “mind”

- ❑ When attributing human-like and mind-like attributes to LLMs, we can ask why and how it generates what it generates.
- ❑ Example techniques
 - Chain of thought (CoT)
 - Self-explanation

CoT for interpretability

- ❑ The “Thought Process” of an LLM?
- ❑ We see the steps an LLM is “following” to perform the tasks
- ❑ Is this a genuine reflection of an internal step-by-step computation, or is the LLM just good at generating plausible-sounding reasoning steps based on patterns in its training data?

Self-explanation

- ❑ Closely related to CoT
- ❑ Asking LLMs to explain and justify their answers

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Faithfulness and Plausibility

- ❑ **Faithfulness**: Does the generated CoT or self-explanation accurately (**faithfully**) reflect the actual internal process the model used to arrive at the answer?
- ❑ Or is it a **plausible** post-hoc rationalization? (LLMs are good at generating plausible text).
- ❑ This is a major challenge in interpretability
 - Aligning external explanations with internal computations.
- ❑ How do humans generate explanations? When are our explanations post-hoc rationalizations vs. true reflections of our thought process? Can studying LLM self-explanation (and its flaws) inform cognitive science, and vice-versa?

Understanding internal mechanisms

Inside the "Black Box"

- ❑ Transformers, Attention Mechanisms (intuitively: “paying attention” to relevant input words).
- ❑ Fundamental training goal: Predict the next word/token.

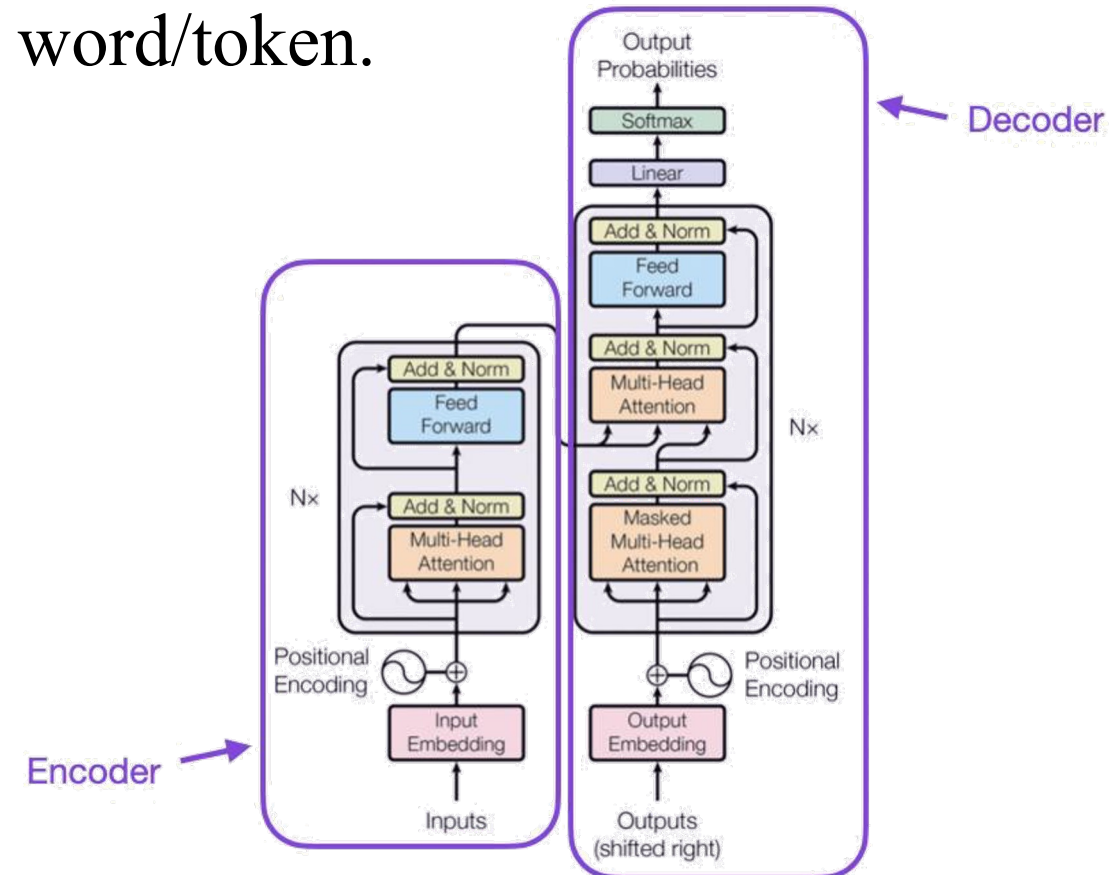


Figure 1: The Transformer - model architecture.

LLMs are implicitly learning other things

- ❑ The simple objective + massive datasets + huge model parameters = Emergent Complexity.
- ❑ **Implicit learning linguistic rules**
 - LLMs aren't explicitly taught grammar rules.
 - They implicitly learn syntax, semantics, and pragmatic patterns from the statistical regularities in the vast corpus.

Why Look Inside?

- ❑ **Trust & Reliability:** Can we depend on their outputs?
- ❑ **Safety & Security:** Identifying and mitigating harmful behaviors (bias, misinformation).
- ❑ **Debugging & Improvement:** Understanding failures to build better models.
- ❑ **Scientific Understanding:** How do these systems actually learn and represent information?
- ❑ **Fairness:** Detecting and addressing biases learned from data.

White-Box Interpretability Techniques

- ❑ **Neuron Activations:** What do individual units respond to?
- ❑ **Attention Visualization:** Where is the model “looking”?
- ❑ **Probing Classifiers:** What information is encoded in representations?
- ❑ **Feature Attribution:** Which inputs drove the output?
- ❑ **Mechanistic Interpretability:** How do components interact to perform computations? (e.g., Circuits - Olah et al.)

Technique 1: Neuron Activations (1)

Region Neurons



USA



Europe



India



West Africa?

Show 3 more neurons.

These neurons respond to content associated with with a geographic region, with neurons ranging in scope from entire hemispheres to individual cities. Some of these neurons partially respond to ethnicity. See [Region Neurons](#) for detailed discussion.

Person Neurons



Donald Trump



Elvis Presley



Lady Gaga



Ariana Grande

Show 1 more neuron.

These neurons respond to content associated with with a specific person. See [Person Neurons](#) for detailed discussion.

Religion Neurons



Judaism



Hinduism



Catholicism



Bible

Show 2 more neurons.

These neurons respond to features associated with a specific religion, such as symbols, iconography, buildings, and texts.

Person Trait Neurons



teenage



elderly



female



male

Show 4 more neurons.

These neurons detect gender¹⁰ and age, as well as facial features like mustaches. (Ethnicity tends to be represented by regional neurons.)

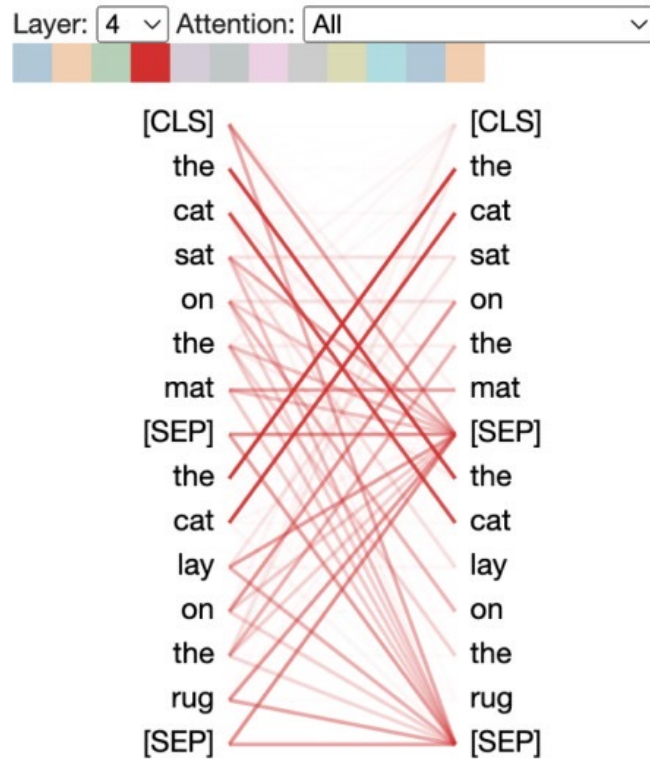
Technique 1: Neuron Activations (2)

- ❑ What Are "Neurons" Doing? Analyzing Activations
- ❑ Idea: Identify what specific concepts, words, or features cause individual "neurons" to activate strongly.
- ❑ E.g., "Is there a 'sentiment neuron'?" or a neuron for "cat-related concepts"?
- ❑ Cognitive Neuroscience Link: Analogous to studying brain regions (fMRI) or single neurons responding to specific stimuli (e.g., faces, places).

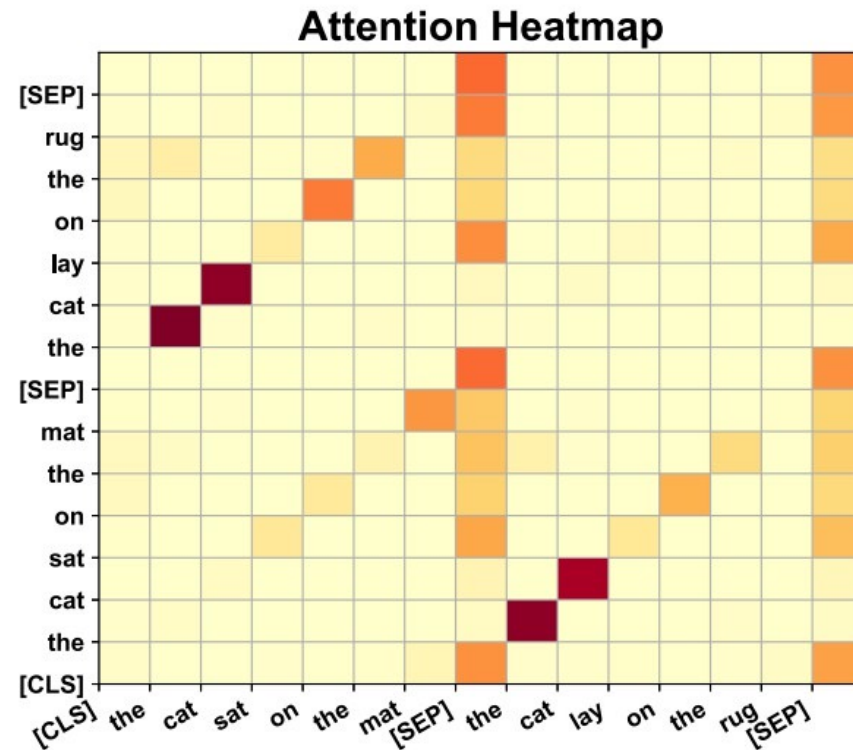
Technique 2: Attention Visualization

- ❑ Where is the Model "Looking"? Visualizing Attention
- ❑ Attention mechanisms weigh the importance of different input tokens when producing an output token.
- ❑ Visualizing these weights shows what the model “focuses on.”
- ❑ **Example:** In translation, a word in the output should attend to corresponding words in the input.

Attention visualization



(a) Bipartite Graph



(b) Heatmap

Fig. 4. Bipartite graph attention representation and heatmap for attention matrix.

<https://dl.acm.org/doi/pdf/10.1145/3639372>

Technique 3: Probing Classifiers (1)

- ❑ What information is encoded? Probing Representations
- ❑ Train **simple** models (“**probes**”) to predict specific properties from LLM's internal representations (activations/embeddings). The representation is frozen and only the simple model is trained.
 - If a probe works well, the information is likely encoded in that layer.
 - By forcing the probe to be simple (like a linear model), it can only succeed if the information is **already easily accessible and linearly separable** within the LLM's representation.

Technique 3: Probing Classifiers (2)

- Probing research has revealed that LLMs generally process language in a structural hierarchy, remarkably similar to traditional linguistics:
 - **Lower Layers (The Foundation):** Probe accurately for surface-level information and morphology (e.g., spelling patterns, token boundaries, parts of speech).
 - **Middle Layers (The Grammar):** Probe highly accurately for syntactic properties (e.g., tracking the subject-verb agreement, parsing tree structures, and determining sentence tense).
 - **Higher Layers (The Meaning):** Probe successfully for semantic and factual knowledge (e.g., sentiment, real-world facts, and context-dependent meanings like the "Theory of Mind" tracking discussed earlier).

Technique 4: Feature Attribution

- ❑ Which inputs mattered most?
- ❑ Methods like LIME, SHAP, Integrated Gradients.
- ❑ Assign an “**importance**” score to each input token for a given output.
- ❑ Helps understand why a specific prediction was made based on the input.

Example categories

- ❑ Gradient-based Methods (e.g., Integrated Gradients):
 - Idea: How much does the output change if an input token changes slightly?
 - Uses the gradient of the output with respect to the input tokens.

- ❑ Perturbation-based Methods (e.g., LIME):
 - Idea: How does the output change if we remove or alter parts of the input?

Limitations of Attribution Methods

- ❑ **Correlation, Not Necessarily Causation:** Highlights what the model associates with the output, but not always how it computes it.
- ❑ **Faithfulness:** Does the explanation truly reflect the model's internal reasoning process?
- ❑ **Focus on Input-Output:** Often treat the model as a black box.
 - Don't fully explain internal interactions between features or complex reasoning steps.
- ❑ **Local Explanations:** Many methods explain individual predictions, not global model behavior.

Technique 5: Mechanistic Interpretability (MI)

- ❑ Attribution tells us what inputs were important.
- ❑ Neuron analysis and probing tell us what information might be encoded where.
- ❑ But how do these pieces combine? How are specific computations or algorithms implemented by the network?
- ❑ This leads to **Mechanistic Interpretability**.
- ❑ The “circuits” metaphor (Olah et al.) aims to uncover these computational pathways. (<https://distill.pub/2020/circuits/>)

MI Philosophy: Understanding by Building & Intervention

- ❑ Feynman's Blackboard: "What I cannot create, I do not understand."
 - MI often aims to build simplified models of how specific circuits work. If the model matches the LLM's behavior, it aids understanding.
- ❑ The Power of Intervention (Causal Interventions):
 - Passive observation (correlations) is not enough.
 - Actively intervening in the model (e.g., ablating components, patching activations) and observing causal effects on behavior is key to testing hypotheses about function.
 - "If I break X, does Y behavior change as predicted?"
- ❑ This iterative process of hypothesis, intervention, and observation drives MI.

MI: The Goal - Reverse Engineering Neural Networks

- ❑ **Goal:** To understand the internal mechanisms and algorithms learned by LLMs, often at the level of individual neurons or small groups (circuits).
- ❑ Moving from correlations (what activates with what) to causal understanding (what component causes what behavior).
- ❑ Analogy: Not just knowing which spark plug is firing, but how the whole engine works.
- ❑ Key Idea: Models learn meaningful "internal algorithms" to perform tasks.

ML: Key Concepts & Techniques

❑ Circuit Analysis:

- Identifying specific "subgraphs" or "circuits" of neurons responsible for particular functions (e.g., detecting repeated text, indirect object identification).

❑ Activation Patching / Causal Tracing:

- Intervene: Run the model on a clean input, then "patch in" activations from a different (corrupted or specific) input into a specific location (e.g., an attention head, an MLP layer).
- Observe: Does this specific patched activation causally restore or change the behavior on the clean input? Helps isolate the role of components.

MI: Challenges

- ❑ **Scalability:** Current techniques are often manual and painstaking, mostly applied to smaller models or specific phenomena in larger ones.
- ❑ **Complexity & Polysemanticity:** Neurons can be involved in multiple unrelated functions. Circuits can be complex and distributed.
- ❑ **Superposition:** Concepts might be represented in a distributed way across many neurons, making individual components hard to interpret.